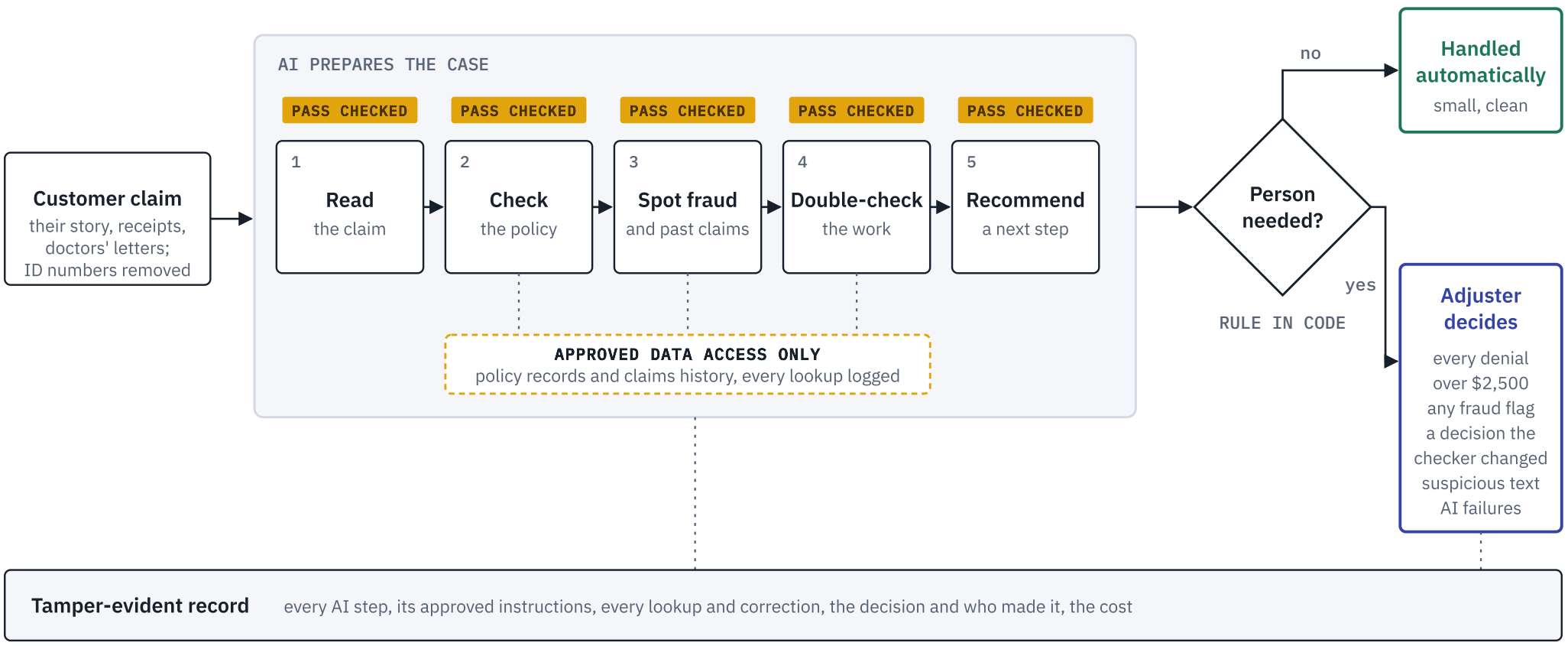
AI that prepares insurance claims, with people making the calls that matter

hall-monitor shows how a regulated firm can use AI agents to speed up claims work **and still answer the regulator's questions**: what each AI was allowed to do, who approved it, what records it looked at, and who made the final decision. The rules live in software, not in a policy binder.

HOW A CLAIM MOVES THROUGH IT



Each AI step has a **hall pass**: a registered, approved permit naming its owner, risk level, the AI model it may use and the records it may look up. A changed permit or changed core instructions are refused until a new approval is committed (not yet covered: the policy wording, the look-up descriptions and the text sending corrections back). Before any AI sees a claim, ID numbers are removed and text that tries to instruct the AI is flagged. The AI gathers evidence and proposes; it never denies or pays a claim on its own. Step 4 is a second AI that checks the others' work against the documents and sends mistakes back. The rule that decides when a person must step in is ordinary software, so the AI can't argue its way past it.

WHAT IT SHOWS SO FAR

0	**Claims that needed a person went through automatically**, in 204 test runs
84/84	**Planted mistakes caught by the double-check**, with 1 false alarm in 34 clean claims
98%	**Coverage decisions right**: amount, outcome and every deciding policy rule cited
0	**Invented facts.** Every extracted value and cited policy rule traced back to a source
3.9¢	**AI cost per claim**, about 21 seconds of processing

WHAT A REGULATOR WILL ASK

What is this AI allowed to do?

Built

Its permit lists the model, the records and the lookups it may use, for this claim only. Anything else is refused and logged.

Who approved it?

Built

Every permit names an owner, and each version and its core instructions need a committed approval to run (the policy wording, look-up descriptions and correction text aren't covered yet). Who may approve is set by repository controls, not yet enforced.

Who makes the big calls?

Built

Denials, large payouts, fraud, and a decision the checking AI changed or still disputes wait for an adjuster, within their authority limit.

What happens to personal data?

Built

ID numbers are removed from claim text before the AI sees it; no claim narrative goes into logs or traces.

Is it accurate and fair?

Partly

Scored on 34 test claims, on earlier versions, misses published. Fairness and attack tests are built; their live results are next. An automated attacker got harm through in 8 of 18 attacks on the complaint desk (published, not yet fixed).

Can an AI that picks its own steps skip work?

Partly

A supervisor AI now picks each next step in all three examples, but sees only codes, can't choose the outcome and is checked by software at every step. Offline, no claim went through automatically that the fixed order sent to a person, and no complaint or leave request got less review; it hasn't run live yet.

What stops AI costs running away?

Built

Approved budgets per case and caps per day and month for each kind of work, changed only by a recorded approval. A live case that would go over goes to a person with no AI call.

Can you prove it months later?

Built

A hash-chained record shows any change. Keep a copy of its latest hash elsewhere to catch a full rewrite.

ROADMAP

1-3	Permits, five AI steps including a double-check, published scores
4-5	Approved data access; adjuster approval tied to the evidence they were shown
6	Tamper-evident record of each claim's AI steps and decision
7	Cost per claim; cheaper models for small claims (off until tested)
8	Privacy, attack defences, fairness and attack tests (results next)
9	Mapped to NIST AI RMF and FINOS; quality gate on every change
10	Decision records, board brief, review package (outside reviewers invited)
11	Hardening: signed adjuster decisions, evidence checked before a decision, data limits enforced at run time
12	Phase B so far: tests of every AI step giving wrong answers, an owner's stop switch, more ID numbers removed (not yet scored live)
13	Test harness for vendor guardrails, gateways and observability; limits as policy code
16-20	HR leave: eligibility rules in code, two AI steps, signed HR sign-off within the role the rules name, a request form (not run live)
21	A complaint desk on real public CFPB complaints: built never to reply to anyone; code sets who handles each complaint at minimum (scripted runs only; the form is proposed)
22-24	Started: access rules for AIs handing work to helpers, an automated attacker, failure drills on many claims at once
25	A supervisor AI now picks the steps in all three examples; code checks every step, holds a budget per case and alone decides the outcome (tested offline only; the scores above are from the older fixed order)
26	Cost governance: approved budgets and daily and monthly caps per workflow (done); caps per AI step, sign-off before costly live runs and a cost page next

A reference implementation, not a production system. Claims, policies and people are fictional; the complaint desk reads real public complaints and contacts no one.

Scores: 34 test claims × 3 runs plus a planted-mistake test, November 2026; they predate milestones 4-11 and Phase B (including stricter routing checks and the kill switch), whose live scores are next. A small sample: it shows the kinds of mistakes, not reliable rates.

github.com/keeplearnin/hall-monitor